

Samadhini Perera

Computer Engineering Undergraduate · Internship Applicant

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Objective

Third-year Computer Engineering undergraduate at the University of Peradeniya seeking an internship to gain real-world industry exposure and broaden technical skills across software engineering, AI/ML, and cloud development. Passionate about building scalable full-stack systems and applying computational methods to real-world engineering problems. Looking to contribute meaningfully to a collaborative, innovation-driven team.

Education

BSc Engineering (Hons) in Computer Engineering

University of Peradeniya, Faculty of Engineering

Jun 2023 – Present (Expected: 2027)

▶ Currently in Year 3; Department of Computer Engineering

▶ Key modules: Data Structures & Algorithms, Software Engineering, Computer Networks, Database Systems, Operating Systems, Embedded Systems, Machine Learning

Technical Skills

Languages	Python	Java	C	JavaScript	SQL	HTML/CSS
Frontend	React 18	Tailwind CSS	React Router	Supabase Client		
Backend	Node.js	Express.js	REST APIs	Swagger	Google OAuth	
Databases	PostgreSQL	Supabase				
AI / ML	Machine Learning	LLMs	Prompt Engineering	Agent-Based Modelling		
DevOps & Tools	Docker	Git	GitHub	Vercel	Render	
Engineering	AutoCAD	Microsoft Excel	Research Writing			

Projects

Event Management Dashboard (Full-Stack)

Stack: React 18 · Node.js · Express · Supabase (PostgreSQL) · Tailwind CSS · Vercel · Google OAuth

Nov 2024 – Nov 2025

▶ Architected a production-ready platform with **separated frontend and backend repositories**, following microservices principles for independent deployment and version control

▶ Developed **15+ reusable React components** with real-time Supabase subscriptions, mobile-first responsive design, cookie-based state persistence, and role-based protected routes

▶ Built **20+ RESTful API endpoints** with interactive Swagger documentation, role-based middleware authentication, input validation, and secure CORS configuration

▶ Features: smart event browsing with filters, live attendance tracking, rating system, CRUD comments, organiser analytics, expense tracking, committee management, dual authentication (Email + Google OAuth)

▶ Live demo: [peraverse.vercel.app](#)

ABM Flood Simulation — Kelani River Basin

Stack: Python · Mesa 2.x · SciPy · NumPy · Plotly Dash · ArcGIS Pro · SRTM DEM

2025 – 2026

- ▶ Extended the Mesa ABM framework with a **custom 3D MeshSpace class** using Delaunay triangulation — a 3D terrain capability absent from all existing open-source ABM platforms
- ▶ Processed **129 GIS-extracted coordinates** (elevation 17–1,601 m, WGS 1984 UTM Zone 44N) generating 248 triangular faces over the Kithulgala and Hanwella study reaches
- ▶ Implemented per-agent hydrodynamic approximation: Manning back-calculated hydraulic depth, overland-flow velocity, and Froude number for flow-regime classification
- ▶ Achieved **O(N) linear time complexity** — 100 agents simulated in 0.06 s versus 2–10 minutes for conventional HEC-RAS, enabling rapid operational flood routing
- ▶ Built an interactive **Plotly Dash dashboard** with 3D animated terrain visualisation and real-time rainfall/runoff sliders; paper prepared for IEEE conference submission

Task Manager App

Stack: Node.js · Express · React · GitHub

2025

- ▶ Full-stack task management application with CRUD operations and RESTful API design; [view on GitHub](#)

Experience

Club Member — Toastmasters International

Peradeniya University Toastmasters Club

Sep 2025 – Present (9 months)

- ▶ Actively developing **public speaking, structured communication, and leadership skills** through prepared speeches, table topics, and peer evaluation in a supportive environment
- ▶ Builds confidence articulating complex technical ideas clearly — directly applicable to client-facing presentations and cross-functional team collaboration in industry

Casual Instructor — Department of Computer Engineering

University of Peradeniya

Nov 2024 – Present (2 years)

- ▶ Instructed **four undergraduate modules**: 2nd Year Project (supervision), Computer Communication Networks, Network Analysis (Electrical), and Programming Methodology, serving students through laboratory sessions
- ▶ **graded assessments** including lab assignments, quizzes, and project evaluations, and mentoring sessions to support student learning

Certifications & Courses (17 completed)

▶ Machine Learning for Beginners — Sololearn	Jan 2026	▶ Introduction to Java — Sololearn	Apr 2025
▶ Introduction to LLMs — Sololearn	Jan 2026	▶ Communication Masterclass — Loku Business	Apr 2025
▶ Prompt Engineering — Sololearn	Jan 2026	▶ C Intermediate — Sololearn	Dec 2024
▶ Agile Project Mgmt in ICT — Univ. of Moratuwa	Dec 2025	▶ HackerRank Achievement — Python, Java, SQL, C, DSA	May 2024
▶ Server-Side Web Programming — Univ. of Moratuwa	Oct 2025	▶ Python for Beginners — Univ. of Moratuwa	May 2024
▶ Corporate & Grooming Etiquettes — Loku Business	Oct 2025	▶ Web Design for Beginners — Univ. of Moratuwa	Jun 2024
▶ Front-End Web Development — Univ. of Moratuwa	Apr 2025	▶ Create Your First Python Program — Coursera/UST	Mar 2024
▶ Foundations of Project Mgmt — Univ. of Moratuwa	Apr 2025	▶ Introduction to C — Sololearn	
▶ Java Intermediate — Sololearn	Apr 2025		

Additional Information

Languages	Sinhala (Native), English (Professional Working Proficiency)
Interests	Painting & Digital Art, Badminton, Yoga
Internship	Seeking opportunity to broaden skills and gain industry exposure ; motivated by learning rather than compensation